

On equations defining space curves

Baron Church

Thm. Let $C \subset \mathbb{P}^n$ be an integral curve of deg d . If C is non-deg ($C \not\subset H$), then C is cut out by equations of deg $\leq d+2-n$.

§. Mumford regularity

Def $\mathcal{F}$ coherent sheaf on $\mathbb{P}^n$ is m -regular if $H^i(\mathbb{P}^n, \mathcal{F}(m-i)) = 0, i > 0$.

Thm (1) any coherent $\mathcal{F}$ is m -regular for $m \gg 0$.

(2) m -regular $\Rightarrow (m+1)$ -regular.

(3) 0 -regular $\Rightarrow$ generated by global sections

(4) m -reg. $\Rightarrow H^0(\mathcal{O}(k)) \otimes H^0(\mathcal{F}(m)) \rightarrow H^0(\mathcal{F}(m+k)), \forall k \geq 0$

(5) $\mathcal{F} \subset \mathcal{O}^{\oplus r}$, then $\mathcal{F}$ is $h(n, r, p(\mathcal{F}))$ -regular.

Use this to prove that $\text{Quot}_r \hookrightarrow \text{Gr}$

$$\{\mathcal{O}^{\oplus r} \twoheadrightarrow \mathcal{F}\}$$

$$\begin{array}{ccc} \mathcal{G} & \rightarrow & \mathcal{O}^{\oplus r} \twoheadrightarrow \mathcal{F} \\ & \searrow & \uparrow \\ & & \mathbb{P}_S^n \end{array} \quad \text{Hart over } S$$

if $\mathcal{G}$ is m -regular, then

$$\pi_* \mathcal{O}^{\oplus r}(m+1) \twoheadrightarrow \pi_* \mathcal{F}(m+1) \text{ is surj. \&}$$

$$\begin{array}{ccc} \text{trivial bundle} & & \text{loc. free} \\ \text{of rk } \binom{n+m+1}{n} r & & \text{rk} = p_{\mathcal{F}}(m+1) \end{array}$$

$$m \text{ is bounded by (5)} \quad \text{Quot}_r \hookrightarrow \text{Gr}\left(\binom{n+m+1}{n} r, p_{\mathcal{F}}(m+1)\right)$$

Thm. $C \subset \mathbb{P}^n$ be a non-deg. integral curve of $\deg = d$. C is $(d+2-n)$ -regular (i.e. I_C is)

Cor. $X \subset \mathbb{P}^n$ is a general hypersurf of $\deg d$. If $d \geq n+1 + \frac{n-k}{e}$ and $e \leq d+r-1$, then X contains finitely many curves of $\deg = e$ spanning a $\mathbb{P}^r$.

Cor. $d=5, n=4$ says that X general has only finitely many rat'l curves of $\deg \leq 7$.

Idea. $\mathcal{X} \subset M_e \times \mathbb{P}$ compute $\dim$ of fiber ($\mathcal{X} \rightarrow M_e$)
 $\ker \left(H^0(\mathcal{O}(d)) \rightarrow H^0(\mathcal{O}_C(d)) \right)$
 $\rightarrow H^1(I_C(d)) \stackrel{if}{=} 0$

§ Berlinson spectral sequence

find a resolution of I_Δ of $\Delta \subset \mathbb{P}^n \times \mathbb{P}^n$

$$\begin{array}{c}
 0 \\
 \downarrow \\
 \pi_2^* \Omega_{\mathbb{P}^n}(1) \\
 \downarrow \\
 0 \rightarrow \pi_1^* \Omega_{\mathbb{P}^n}(1) \rightarrow \mathcal{O}_{\mathbb{P}^n \times \mathbb{P}^n}^{\oplus (n+1)} \xrightarrow{\pi_1^* s} \pi_1^* \mathcal{O}(1) \rightarrow 0 \\
 \searrow \psi \quad \downarrow \pi_2^* s \\
 \pi_2^* \mathcal{O}(1) \\
 \downarrow \\
 0
 \end{array}$$

$$\Delta = V(\psi)$$

$$M = \pi_1^* \Omega_{\mathbb{P}^n}(1)$$

$$0 \rightarrow \left[\Lambda^n M \otimes \pi_2^* \mathcal{O}(-n) \rightarrow \dots \rightarrow M \otimes \pi_2^* \mathcal{O}(-1) \right] \rightarrow I_\Delta \rightarrow 0 \quad \text{is exact}$$

Any coherent $\mathcal{F}$ satisfies $\mathcal{F} = \pi_{2*} \left((\pi_1^* \mathcal{F}) \otimes I_\Delta \right)$

$$\begin{aligned} E_1^{p,q} &= R^q \pi_{2*} \left(\pi_1^* (\mathcal{F} \otimes \Omega_{\mathbb{P}^n}^{-p}(-p)) \otimes \pi_2^* \mathcal{O}(p) \right) \Rightarrow \begin{cases} \mathcal{F} & p=q=0 \\ 0 & \text{else} \end{cases} \\ &= R^q \pi_{2*} \left(\pi_1^* (\mathcal{F} \otimes \Omega_{\mathbb{P}^n}^{-p}(-p)) \right) \otimes \mathcal{O}(p) \\ &= H^q(\mathcal{F} \otimes \Omega_{\mathbb{P}^n}^{-p}(-p)) \otimes \mathcal{O}(p) \end{aligned}$$

Lemma: Suppose $\dim \text{Supp } \mathcal{F} = 1$ and $H^1(\mathbb{P}^n, \mathcal{F} \otimes \Omega^2(2)) = 0$, then

$$H^0(\mathbb{P}^n, \mathcal{F} \otimes \Omega^1(1)) \otimes \mathcal{O}(-1) \rightarrow H^0(\mathbb{P}^n, \mathcal{F}) \otimes \mathcal{O} \rightarrow \mathcal{F} \rightarrow 0$$

$$\dim \Rightarrow H^{>1}(\mathbb{P}^n, \mathcal{F} \otimes \Omega^p(p)) = 0.$$

§ Eagon - Northcott complex

Thm: $u: E \rightarrow F$ generically surj, $\exists$ cplx

$$0 \rightarrow \Lambda^e E \otimes S^{e-b}(F)^\vee \rightarrow \Lambda^{e-1} E \otimes S^{e-b-1}(F)^\vee \rightarrow \dots \rightarrow \Lambda^b E \xrightarrow{\Lambda^b u} \Lambda^b F \rightarrow 0$$

exact away from $\text{supp } \text{coker } u$.

Construction. $S = S^*(F)$ graded ring

[I think it's just the sym power of the cplx $[E \rightarrow F]$]

§ Proof of Thm.

$C \subset \mathbb{P}^n$ be an integral curve of $\deg = d$. $f: \tilde{C} \rightarrow C \subset \mathbb{P}^n$ be normalization.

Set $M = \Omega_{\mathbb{P}^n}(1)$.

Thm Assume A is a l.b. on $\tilde{C}$ s.t. $H^1(\tilde{C}, \Lambda^2 f^* M \otimes A) = 0$, then

C is $\underbrace{h^0(\tilde{C}, A)}_{n_0}$ -regular.

Proof f is finite, so $H^i(\tilde{C}, \mathcal{F}) = H^i(\mathbb{P}^n, f_* \mathcal{F})$

$H^1(\mathbb{P}^n, \Lambda^2 M \otimes f_* A) = 0$ by assumption $\xRightarrow{\text{Beilinson SS}}$

$H^0(\tilde{C}, f^* M \otimes A) \otimes \mathcal{O}(-1) \xrightarrow{\sim} \widehat{H^0(\tilde{C}, A)}^{\otimes n_0} \otimes \mathcal{O} \rightarrow f_* A \rightarrow 0$ exact

EN cplx $E \rightarrow F$

$$\underbrace{\mathcal{O}(-k)^{M_k} \rightarrow \mathcal{O}(-k+1)^{M_{k-1}} \rightarrow \dots \rightarrow \mathcal{O}(-n_0)^{M_{n_0}}}_{C,} \rightarrow \mathcal{O}^M \rightarrow \mathcal{O} \otimes A \rightarrow 0$$

exact away from C : 1-dim'l

$I = \text{im}(\wedge^{n_0} u)$ ~ 0-th Fitting ideal of $f_* A$. $I \subset I_C$

and $\text{supp}(I_C/I)$ 0-dim'l (exactly where $f_* A$ fails to be
loc. free $\subset \text{sing}(C)$)

$H^i(\mathbb{P}^n, I(m)) \rightarrow H^i(\mathbb{P}^n, I_C(m)), i > 0$

suffices to show I is n_0 -regular.

Notice $H^i(C_t(n_0-k)) = 0, \quad i > 0, \quad \text{for } 1 \leq k \leq n.$

if $t > -(n_0-k)$

Lemma Suppose $F_0 \xrightarrow{\epsilon} G \rightarrow 0 \quad 1 \leq k \leq n \quad \epsilon \text{ surj. on } X \text{ } n\text{-dim'l proj.}$

(a) exact away from dim 1

(b) $H^i(F_0) = H^i(F_1) = \dots = H^i(F_{n-k}) = 0$

then $H^i(X, G) = 0$ for $i \geq k$

$\Rightarrow H^i(\mathbb{P}^n, I(n_0-k)) = 0$ for $i \geq k \quad \Rightarrow I$ is n_0 -regular.

Lemma if genus $(\tilde{C}) = g$, then general bb. A of $\deg = g + d + 1 - n$ satisfies

$$(1) \quad H^1(\tilde{C}, \sigma^* \Lambda^2 M \otimes A) = 0,$$

$$(2) \quad h^0(\tilde{C}, A) = d + 2 - n.$$

